ENN581 Robot Motion, Control and Planning · Mock exam 1 — questions

Indicative marks in [brackets] (the real paper's marks are not published). 2 hours. Calculator and printed notes allowed. Write all working.

Mock exam 1 indicative total 64 marks · 2 hours

Q1 · Forward kinematics of a SCARA arm derive calculate 6 marks

A SCARA arm stands on a column of height H = 0.40 m. Joint 1 rotates about the vertical axis of the column. A horizontal link of length a_1 = 0.35 m leads to joint 2, which also rotates about a vertical axis. A second horizontal link a_2 = 0.25 m carries joint 3, a prismatic joint that extends the tool *downwards* by d_3. The tool frame {T} has z_T pointing down and x_T along link 2.

(a) Write an elementary transform sequence for ${}^0\mathbf{T}_T$ and a proximal DH table for the arm. [3]
(b) Evaluate ${}^0\mathbf{T}_T$ at $\theta_1 = 30°$, $\theta_2 = -60°$, d_3 = 0.10 m. [3]

Q2 · Inverse kinematics and workspace of the SCARA derive sketch 8 marks

For the arm of Q1: (a) derive θ_1, θ_2, d_3 for a desired tool position (x, y, z) and state how many solutions exist [4]; (b) evaluate them for (0.45, 0.20, 0.25) m [2]; (c) sketch the workspace if $0 \le d_3 \le 0.20$ m and the revolute joints are unlimited [2].

Q3 · Mobility calculate explain 6 marks

Find the mobility of: (a) a planar five-bar linkage (five links including ground, five revolute joints) [1]; (b) a 3-RPS parallel platform (each leg: revolute at the base, prismatic, spherical at the platform) [2]; (c) a 3-SPS platform (spherical joints at both ends of each prismatic leg). Explain any correction you apply [3].

Q4 · Two cubic segments through a via point derive calculate 6 marks

A joint starts at rest at $\theta_0 = 0°$, passes through $\theta_v = 40°$ and stops at $\theta_g = 30°$. Each segment lasts 1 s and the acceleration is continuous at the via point. (a) List the conditions [2]. (b) Find all eight coefficients [3]. (c) What is the joint speed at the via point? [1]

Q5 · Jacobian, singularity and static force derive calculate explain 7 marks

A planar 2R arm has $L_1 = 0.5$ m, $L_2 = 0.4$ m. (a) Derive its 2×2 position Jacobian and its singular configurations [3]. (b) At $q = (30°, 60°)$, find the joint torques needed to hold a tip force $F = (0, -10)$ N, and explain why τ_2 comes out as it does [2]. (c) Find $\dot{\mathbf{q}}$ for a tip velocity (0.1, 0) m/s at that pose [2].

Q6 · Equations of motion of an RP arm derive calculate 8 marks

An RP arm moves in a vertical plane. Joint 1 rotates the arm by θ from the horizontal. Joint 2 slides a point mass m along the arm to distance d from joint 1. The arm itself is massless; gravity acts along −y. (a) Derive the equations of motion by Lagrange and identify M, C (velocity terms) and g [6]. (b) Evaluate τ_1 and f_2 for m = 2 kg at θ = 30°, d = 0.6 m, $\dot{\theta}$ = 0.5 rad/s, $\dot{d}$ = 0.2 m/s, $\ddot{\theta}$ = 1.0 rad/s², $\ddot{d}$ = −0.3 m/s² [2].

Q7 · Partitioned control of one joint derive calculate 7 marks

A joint obeys $m\ddot{x} + b\dot{x} + kx = f$ with m = 2 kg, b = 0.5 N·s/m, k = 4 N/m. (a) Write a partitioned control law for trajectory following and the closed-loop error equation [3]. (b) Choose k_p, k_v for critical damping with ω_n = 8 rad/s [1]. (c) Find the steady-state error under a constant 5 N disturbance, and say how to remove it [2]. (d) Compute f at x = 0.10 m, $\dot{x}$ = −0.20 m/s when x_d = 0.15 m, $\dot{x}_d = \ddot{x}_d = 0$ [1].

Q8 · Differential-drive robot derive calculate explain 6 marks

Wheels of radius r = 0.08 m sit at (0, ±a) with a = 0.20 m (left wheel 1 at +a). (a) From the generalised wheel model, derive the chassis velocity from the wheel speeds [2]. (b) For $\dot{\boldsymbol{\varphi}}$ = (5, 7.5) rad/s find u, $\dot{\theta}$ and the ICR [2]. (c) Find the wheel speeds for u = 0.3 m/s, $\dot{\theta}$ = −0.5 rad/s [1]. (d) Write the robot's non-holonomic constraint in the inertial frame [1].

Q9 · A* by hand calculate explain 6 marks

Undirected edges with costs: S–A 2, S–B 5, A–C 4, A–D 7, B–D 3, C–G 6, D–G 2. Heuristic h: S 7, A 6, B 4, C 5, D 2, G 0. (a) Run A* from S to G; give the expansion order with g, h, f, and the path and cost [4]. (b) Is h admissible? Consistent? [2]

Q10 · Wheeled-robot dynamics (short) explain 4 marks

The chassis frame {C} moves with velocity (u, v) and turns at $\dot{\theta}$. Show that the acceleration of the chassis origin, written in {C}, is $(\dot{u} - v\dot{\theta},\ \dot{v} + u\dot{\theta})$, and explain why a robot driving in a circle at constant u needs a sideways force even though v = 0.